

C program Lab Task

Functions

- Write a function to calculate the square of a number.
- Write a function to check if a number is even or odd.
- Real-time: Create a function `calculateBill()` that takes quantity and price, and returns the total bill.



Arrays

- Input 10 numbers into an array and print them in reverse order.
- Find the largest and smallest element in an array.
- Real-time: Store daily temperatures for a week and calculate the average.



2D Arrays

- Input a 3×3 matrix and print it.
- Find the sum of all elements in a 2D array.
- Real-time: Store marks of 3 students in 3 subjects and print each student's total.



Strings

- Input a string and count the number of vowels.
- Reverse a string without using `strrev()`.
- Real-time: Input a student's name and roll number, then print a formatted ID card.



Pointers

- Demonstrate pointer basics: print the value and address of a variable.
- Swap two numbers using pointers.
- Real-time: Use pointers to dynamically allocate memory for storing 5 student marks.



Structures

- Define a structure `Student` with roll, name, and marks. Input and display details.
- Store details of 5 students in an array of structures.
- Real-time: Find the student with the highest marks using structures.



Union

- Define a union with `int`, `float`, and `char`. Demonstrate how only one member can hold a value at a time.

- Real-time: Use a union to store either a student's roll number or name depending on context (memory saving).

File Handling

- Write a program to create a file and write "Hello World" into it.
- Read contents of a file and display them on the screen.
- Real-time: Store student marks in a file and read them back for processing.